

TP 6: Apache Yarn

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Année universitaire :

- 2025–2026

Prérequis et vérification de l'environnement

1. Verification de java et Hadoop

```
mohammed@ROG-Strix:~/HADOOP_REPO/yarn
> java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment (build 1.8.0_482-8u482-ga~us1-0ubuntu1~24.04-b08)
OpenJDK 64-Bit Server VM (build 25.482-b08, mixed mode)
> hadoop version
Hadoop 2.7.1
Subversion https://git-wip-us.apache.org/repos/asf/hadoop.git -r 15ecc87ccf4a0228f35af08fc56de536e6ce657a
Compiled by jenkins on 2015-06-29T06:04Z
Compiled with protoc 2.5.0
From source with checksum fc0a1a23fc1868e4d5ee7fa2b28a58a
This command was run using /home/mohammed/hadoop/share/hadoop/common/hadoop-common-2.7.1.jar
> echo $HADOOP_COMMON_HOME
/home/mohammed/hadoop
> echo $JAVA_HOME
/usr/lib/jvm/java-8-openjdk-amd64
yarn >
```

2. Démarrer HDFS et YARN

```
mohammed@ROG-Strix:~/HADOOP_REPO/yarn
> start-dfs.sh
Starting namenodes on [localhost]
localhost: starting namenode, logging to /home/mohammed/hadoop/logs/hadoop-mohammed-namenode-ROG-Strix.out
localhost: starting datanode, logging to /home/mohammed/hadoop/logs/hadoop-mohammed-datanode-ROG-Strix.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /home/mohammed/hadoop/logs/hadoop-mohammed-secondarynamenode-ROG-Strix.out
> start-yarn.sh
starting yarn daemons
starting resourcemanager, logging to /home/mohammed/hadoop/logs/yarn-mohammed-resourcemanager-ROG-Strix.out
localhost: starting nodemanager, logging to /home/mohammed/hadoop/logs/yarn-mohammed-nodemanager-ROG-Strix.out
> jps
12883 NodeManager
12565 ResourceManager
11973 NameNode
13030 Jps
10918
12363 SecondaryNameNode
12140 DataNode
yarn >
```

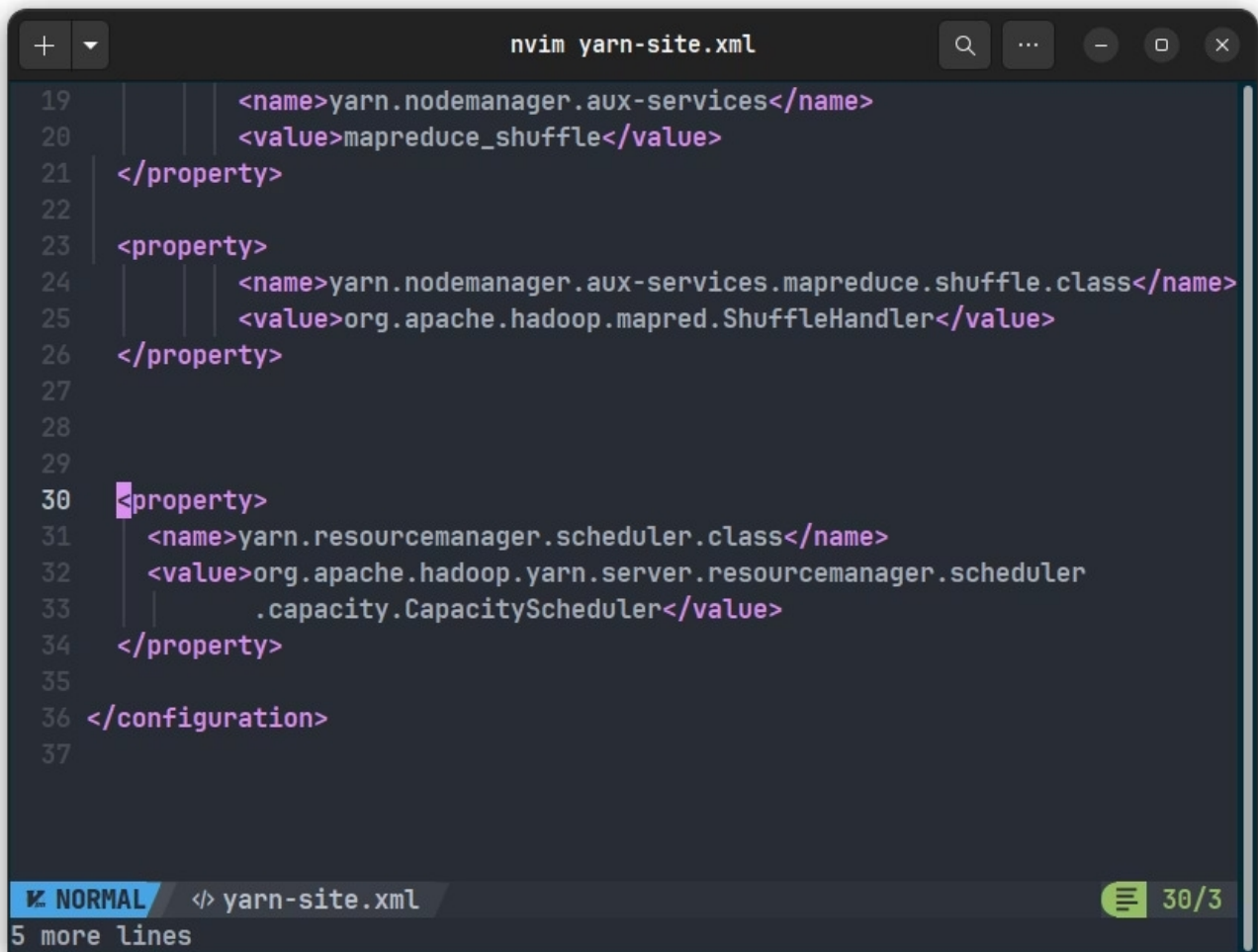
3. Préparer les données de test sur HDFS

```
mohammed@ROG-Strix:~/HADOOP_REPO/yarn
> hdfs dfs -mkdir -p /yarn-demo/input
> echo "hadoop yarn scheduler capacity fair resource management cluster big data" > test.txt
> hdfs dfs -put test.txt /yarn-demo/input/
> hdfs dfs -ls /yarn-demo/input/
Found 1 items
-rw-r--r--  1 mohammed supergroup          74 2026-04-22 11:55 /yarn-demo/input/test.txt
> hdfs dfs -cat /yarn-demo/input/test.txt
hadoop yarn scheduler capacity fair resource management cluster big data
yarn
```

4. Création de fichier capacity-scheduler.xml

```
nvim capacity-scheduler.xml
1 <?xml version="1.0"?>
2 <configuration>
3
4   <!-- Définir les 3 queues au niveau root -->
5   <property>
6     <name>yarn.scheduler.capacity.root.queues</name>
7     <value>engineering,research,default</value>
8   </property>
9
10  <!-- Queue engineering : 50% garanti, 70% max (burst) -->
11  <property>
12    <name>yarn.scheduler.capacity.root.engineering.capacity</name>
13    <value>50</value>
14  </property>
15  <property>
16    <name>yarn.scheduler.capacity.root.engineering.maximum-capacity</name>
17    <value>70</value>
18  </property>
19  <property>
20    <name>yarn.scheduler.capacity.root.engineering.state</name>
21    <value>RUNNING</value>
22  </property>
```

5. Activer la capacity

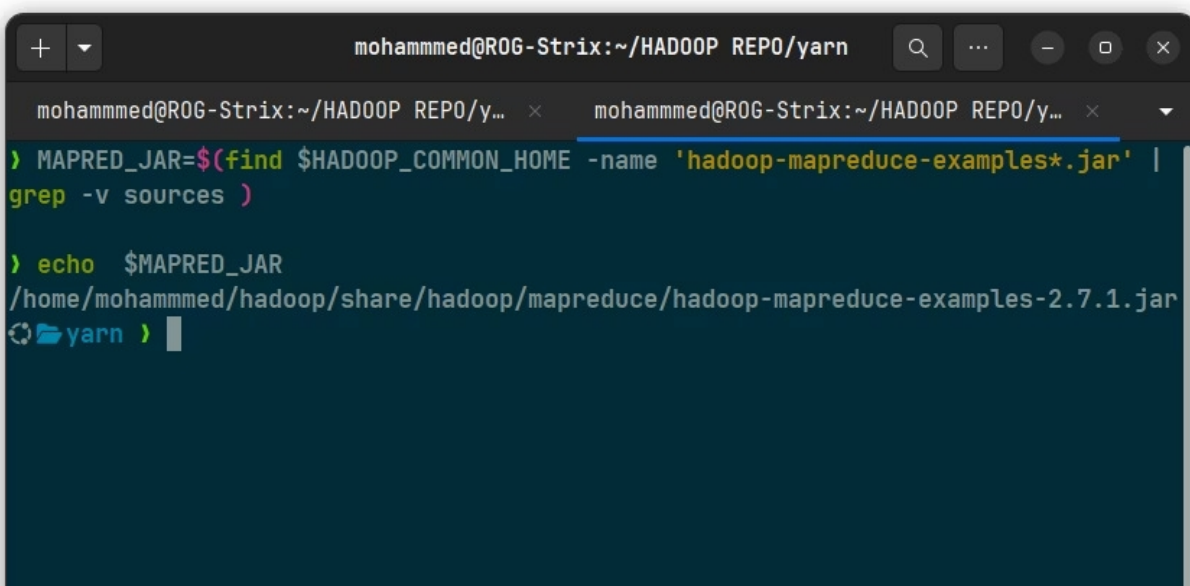


```
nvim yarn-site.xml
19      <name>yarn.nodemanager.aux-services</name>
20      <value>mapreduce_shuffle</value>
21    </property>
22
23    <property>
24      <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
25      <value>org.apache.hadoop.mapred.ShuffleHandler</value>
26    </property>
27
28
29
30  <property>
31    <name>yarn.resourcemanager.scheduler.class</name>
32    <value>org.apache.hadoop.yarn.server.resourcemanager.scheduler
33      .capacity.CapacityScheduler</value>
34  </property>
35
36 </configuration>
37
```

5 more lines

6. Redémarrer Yarn soumettre des jobs

1. Trouver le Jar exemples MapReduce



```
mohammed@ROG-Strix:~/HADOOP_REPO/yarn
> MAPRED_JAR=$(find $HADOOP_COMMON_HOME -name 'hadoop-mapreduce-examples*.jar' |
grep -v sources )
> echo $MAPRED_JAR
/home/mohammed/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-2.7.1.jar
yarn >
```


6. Soumettre des jobs dans les queues

A. soumettre job 1

```
mohammed@ROG-Strix:~/HADOOP_REPO/yarn
> hadoop jar $MAPRED_JAR wordcount \
-Dmapreduce.job.queueName=engineering \
-Dmapreduce.job.name='WC-Engineering' \
/yarn-demo/input /yarn-demo/output-eng &
[2] 28198
yarn 26/04/25 13:25:39 INFO client.RMPProxy: Connecting to ResourceManager
at localhost/127.0.0.1:8032
26/04/25 13:25:40 INFO input.FileInputFormat: Total input paths to process : 1
26/04/25 13:25:40 INFO mapreduce.JobSubmitter: number of splits:1
26/04/25 13:25:40 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_17
77119793773_0002
26/04/25 13:25:41 INFO impl.YarnClientImpl: Submitted application application_17
77119793773_0002
26/04/25 13:25:41 INFO mapreduce.Job: The url to track the job: http://ROG-Strix
:8088/proxy/application_1777119793773_0002/
26/04/25 13:25:41 INFO mapreduce.Job: Running job: job_1777119793773_0002
26/04/25 13:25:45 INFO mapreduce.Job: Job job_1777119793773_0002 running in uber
mode : false
26/04/25 13:25:45 INFO mapreduce.Job: map 0% reduce 0%
26/04/25 13:25:48 INFO mapreduce.Job: map 100% reduce 0%
26/04/25 13:25:52 INFO mapreduce.Job: map 100% reduce 100%
26/04/25 13:25:52 INFO mapreduce.Job: Job job_1777119793773_0002 completed succe
ssfully
26/04/25 13:25:52 INFO mapreduce.Job: Counters: 49
```

B. Soumettre job 2

```
mohammed@ROG-Strix:~/HADOOP_REPO/yarn
> hadoop jar $MAPRED_JAR wordcount \
-Dmapreduce.job.queueName=research \
-Dmapreduce.job.name='WC-Research' \
/yarn-demo/input /yarn-demo/output-res &
[2] 28726
yarn 26/04/25 13:27:04 INFO client.RMPProxy: Connecting to ResourceManager
at localhost/127.0.0.1:8032
26/04/25 13:27:04 INFO input.FileInputFormat: Total input paths to process : 1
26/04/25 13:27:04 INFO mapreduce.JobSubmitter: number of splits:1
26/04/25 13:27:04 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_17
77119793773_0003
26/04/25 13:27:04 INFO impl.YarnClientImpl: Submitted application application_17
77119793773_0003
26/04/25 13:27:04 INFO mapreduce.Job: The url to track the job: http://ROG-Strix
:8088/proxy/application_1777119793773_0003/
26/04/25 13:27:04 INFO mapreduce.Job: Running job: job_1777119793773_0003
26/04/25 13:27:08 INFO mapreduce.Job: Job job_1777119793773_0003 running in uber
mode : false
26/04/25 13:27:08 INFO mapreduce.Job: map 0% reduce 0%
26/04/25 13:27:13 INFO mapreduce.Job: map 100% reduce 0%
26/04/25 13:27:17 INFO mapreduce.Job: map 100% reduce 100%
26/04/25 13:27:17 INFO mapreduce.Job: Job job_1777119793773_0003 completed succe
ssfully
26/04/25 13:27:17 INFO mapreduce.Job: Counters: 49
```

C. voir les apps en cours

```

+  mohammed@ROG-Strix:~/HADOOP_REPO/yarn
mohammed@ROG-Strix:~$ yarn application -list -appStates ALL
26/04/25 13:28:48 INFO client.RMPProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
26/04/25 13:28:48 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Total number of applications (application-types: [] and states: [NEW, NEW_SAVING, SUBMITTED, ACCEPTED, RUNNING, FINISHED, FAILED, KILLED]):2

Application-Id      Application-Name      Application-Type
User                Queue                State                Final-State
Progress            Tracking-URL

application_1777119793773_0003      word count      MAPREDUCE
mohammed      research      FINISHED      SUCCEEDED
100%      http://ROG-Strix:19888/jobhistory/job/job_1777119793773_0003
application_1777119793773_0002      word count      MAPREDUCE
mohammed      engineering      FINISHED      SUCCEEDED
100%      http://ROG-Strix:19888/jobhistory/job/job_1777119793773_0002
yarn
```

D. Sur le site

Cluster

About Nodes Node Labels Applications

NEW NEW_SAVING SUBMITTED ACCEPTED RUNNING FINISHED FAILED KILLED Scheduler

Tools

All Applications

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	V-Cores Used	V-Cores Total	V-Cores Reserved	Active Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
4	0	0	4	0	0 B	8 GB	0 B	0	8	0	1	0	0	0	0

Scheduler Type

Scheduling Resource Type

Minimum Allocation

Maximum Allocation

Capacity Scheduler

[MEMORY]

<memory:1024, vCores:1>

<memory:8192, vCores:8>

Show 20 entries

Search

ID	User	Name	Application Type	Queue	StartTime	FinishTime	State	FinalStatus	Progress	Tracking UI
application_1777119793773_0005	mohammed	QuasiMonteCarlo	MAPREDUCE	research	Sat Apr 25 13:30:20 +0100 2026	Sat Apr 25 13:30:33 +0100 2026	FINISHED	SUCCEEDED		History
application_1777119793773_0004	mohammed	QuasiMonteCarlo	MAPREDUCE	research	Sat Apr 25 13:29:53 +0100 2026	Sat Apr 25 13:30:07 +0100 2026	FINISHED	SUCCEEDED		History
application_1777119793773_0003	mohammed	word count	MAPREDUCE	research	Sat Apr 25 13:27:04 +0100 2026	Sat Apr 25 13:27:15 +0100 2026	FINISHED	SUCCEEDED		History
application_1777119793773_0002	mohammed	word count	MAPREDUCE	engineering	Sat Apr 25 13:26:40 +0100 2026	Sat Apr 25 13:25:51 +0100 2026	FINISHED	SUCCEEDED		History

Showing 1 to 4 of 4 entries

First Previous 1 Next Last

Fair Scheduler avec DRF et Prémption

1. création de fichier fair-scheduler.xml

```
nvim $HADOOP_COMMON_HOME/etc/hadoop/fair-scheduler.xml
1 <allocations>
2   <defaultQueueSchedulingPolicy>drf</defaultQueueSchedulingPolicy>
3
4   <queue name="engineering">
5     <minResources>2048mb,2vcores</minResources>
6     <maxResources>8192mb,8vcores</maxResources>
7     <weight>3.0</weight>
8     <schedulingPolicy>fair</schedulingPolicy>
9   </queue>
10
11   <queue name="research">
12     <minResources>1024mb,1vcores</minResources>
13     <maxResources>6144mb,6vcores</maxResources>
14     <weight>2.0</weight>
15     <schedulingPolicy>drf</schedulingPolicy>
16   </queue>
17
18   <queue name="default">
19     <minResources>512mb,1vcores</minResources>
20     <maxResources>4096mb,4vcores</maxResources>
21     <weight>1.0</weight>
22   </queue>
23 </allocations>
```

2. Activation de le fair scheduler dans yarn-site.xml

```
nvim $HADOOP_CONF_DIR/yarn-site.xml
12
11 <property>
10   <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
9   <value>org.apache.hadoop.mapred.ShuffleHandler</value>
8 </property>
7 <property>
6   <name>yarn.resourcemanager.scheduler.class</name>
5   <value>org.apache.hadoop.yarn.server.resourcemanager.scheduler.fair.FairS
scheduler</value>
4 </property>
3
2 <property>
1   <name>yarn.scheduler.fair.allocation.file</name>
37 <value>/home/mohammed/hadoop/etc/hadoop/fair-scheduler.xml</value>
1
2 </property>
3
4 <property>
5   <name>yarn.scheduler.fair.preemption</name>
6   <value>true</value>
7 </property>
8 </configuration>
```


3. Redemeraer YARN et vérifier les queues

```
mohammed@ROG-Strix:~/ .zsh
mohammed@ROG-Strix:~/ .zsh x mohammed@ROG-Strix:~/HADOOP_REPO/... x
26/04/25 14:19:47 WARN util.NativeCodeLoader: Unable to load native-hadoop libra
ry for your platform... using builtin-java classes where applicable
Queue Information :
Queue Name : root.engineering
State : RUNNING
Capacity : .0%
Current Capacity : .0%
Maximum Capacity : -100.0%
Default Node Label expression :
Accessible Node Labels :
) yarn queue -status research
26/04/25 14:20:13 INFO client.RMProxy: Connecting to ResourceManager at localhos
t/127.0.0.1:8032
26/04/25 14:20:13 WARN util.NativeCodeLoader: Unable to load native-hadoop libra
ry for your platform... using builtin-java classes where applicable
Queue Information :
Queue Name : root.research
State : RUNNING
Capacity : .0%
Current Capacity : .0%
Maximum Capacity : -100.0%
Default Node Label expression :
Accessible Node Labels :
.zsh )
```

Resultat sur le web

NEW,NEW_SAVING,SUBMITTED,ACCEPTED,RUNNING Applications

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	VCores Used	VCores Total	VCores Reserved	Active Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
1	0	0	1	0	0B	8 GB	0B	0	8	0	1	0	0	0	0

User Metrics for dr.who

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Containers Pending	Containers Reserved	Memory Used	Memory Pending	Memory Reserved	VCores Used	VCores Pending	VCores Reserved
0	0	0	0	0	0	0	0B	0B	0B	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Fair Scheduler	[MEMORY, CPU]	<memory:1024, vCores:1>	<memory:8192, vCores:8>

Application Queues

Legend: ☐ Steady Fair Share ☐ Instantaneous Fair Share ☒ Used ☐ Used (over fair share) ☐ Max Capacity

- root
 - root.default 0.0% used
 - root.research 0.0% used
 - root.engineering 0.0% used

Show 20 entries

ID	User	Name	Application Type	Queue	Fair Share	StartTime	FinishTime	State	FinalStatus	Progress	Tracking UI
No data available in table											

Showing 0 to 0 of 0 entries

WhatsApp

Copy of Spark | Gamma

Spotify - Web Player:

Ubuntu setting - Local

NEW, NEW_SAVING, SUBM

localhost:8088/cluster/scheduler?openQueues=root#root.default#root.research#root.engineering

ChatGPTWhatsAppYouTubeFacebookA pink and...Ubuntu | Do...

KILLED

Scheduler

Tools

Fair Scheduler

Application Queues

Legend: Steady Fair ShareInstantaneous Fair ShareUsedUsed (over fair share)Max Capacity

root

root.default

root.research

root.engineering

'root.default' Queue Status

Used Resources: <memory:0, vCores:0>

Num Active Applications: 0

Num Pending Applications: 0

Min Resources: <memory:512, vCores:1>

Max Resources: <memory:4096, vCores:4>

Steady Fair Share: <memory:1365, vCores:1>

Instantaneous Fair Share: <memory:0, vCores:0>

'root.research' Queue Status

Used Resources: <memory:0, vCores:0>

Num Active Applications: 0

Num Pending Applications: 0

Min Resources: <memory:1024, vCores:1>

Max Resources: <memory:6144, vCores:6>

Max Running Applications: 5

Steady Fair Share: <memory:2731, vCores:3>

Instantaneous Fair Share: <memory:0, vCores:0>

'root.engineering' Queue Status

Used Resources: <memory:0, vCores:0>

Num Active Applications: 0

Num Pending Applications: 0

Min Resources: <memory:2048, vCores:2>

Max Resources: <memory:8192, vCores:8>

Max Running Applications: 10

Steady Fair Share: <memory:4096, vCores:4>

Instantaneous Fair Share: <memory:0, vCores:0>

Show 20 entries

Search:

ID	User	Name	Application Type	Queue	Fair Share	StartTime	FinishTime	State	FinalStatus	Progress	Tracking UI
No data available in table											